

Introduction to SQL

Miao Qiao

The University of Auckland



Basic Query Structure

- A typical SQL query has the form:

select $A_1, A_2, \dots, A_n$
from $r_1, r_2, \dots, r_m$
where P

<i>department</i>	<i>instructor</i>	<i>teaches</i>	<i>section</i>	<i>takes</i>	<i>student</i>	<i>course</i>	<i>advisor</i>
<u><i>dept_name</i></u> <i>building</i> <i>budget</i>	<u><i>ID</i></u> <i>name</i> <i>dept_name</i> <i>salary</i>	<u><i>ID</i></u> <u><i>course_id</i></u> <u><i>sec_id</i></u> <u><i>semester</i></u> <u><i>year</i></u>	<u><i>course_id</i></u> <u><i>sec_id</i></u> <u><i>semester</i></u> <u><i>year</i></u> <i>building</i> <i>room_number</i> <i>time_slot_id</i>	<u><i>ID</i></u> <u><i>course_id</i></u> <u><i>sec_id</i></u> <u><i>semester</i></u> <u><i>year</i></u> <i>grade</i>	<u><i>ID</i></u> <i>name</i> <i>dept_name</i> <i>tot_cred</i>	<u><i>course_id</i></u> <i>title</i> <i>dept_name</i> <i>credits</i>	<u><i>s_id</i></u> <u><i>i_id</i></u>

Basic Query Structure (demo)

1. Find the names of all instructors
2. Find the department names of all instructors, and remove duplicates
3. Find the department names of all instructors, not removing duplicates
4. Find all attributes of instructor show the entire instructor table
5. Find a relation that is the same as the *instructor* relation, except that the value of the attribute *salary* is divided by 12
6. Find all instructors in Comp. Sci. dept with salary > 70000
7. Find the names of all instructors who have taught some course and the course_id
8. Find the names of all instructors in the Comp. Sci. department who have taught some course and the course_id
9. Find the names of all instructors who have a higher salary than some instructor in 'Comp. Sci'.

				section				
department	instructor	teaches			takes	student	course	advisor
<u>dept_name</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	<u>course_id</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	
building	name	<u>sec_id</u>	<u>sec_id</u>	<u>sec_id</u>	<u>course_id</u>	name	title	<u>s_id</u>
	dept_name	<u>semester</u>	<u>semester</u>	<u>year</u>	<u>sec_id</u>	dept_name	dept_name	<u>i_id</u>
	salary	<u>year</u>	<u>year</u>	<u>building</u>	<u>semester</u>	tot_cred	credits	
				room_number	<u>year</u>			
				time_slot_id	grade			

Basic Query Structure (demo)

- Find the names of all instructors whose name includes the substring “in”.
- String Operations
 - The operator **like** uses patterns that are described using two special characters:
 - percent (%). The % character matches any substring.
 - underscore (_). The _ character matches any character
- Find the names of all instructors whose name has 4 characters.
- Find the names of all instructors whose name has at least 4 characters.

				section				
department	instructor	teaches			takes	student	course	advisor
<u>dept_name</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	<u>course_id</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	<u>s_id</u>
building	name	<u>sec_id</u>	<u>semester</u>	<u>sec_id</u>	<u>course_id</u>	name	title	<u>i_id</u>
	dept_name	<u>semester</u>	<u>year</u>	<u>building</u>	<u>sec_id</u>	dept_name	dept_name	
	salary	<u>year</u>	<u>time_slot_id</u>	room_number	<u>semester</u>	tot_cred	credits	
					<u>year</u>			
					grade			

Basic Query Structure (demo)

1. List in alphabetic order the names of all instructors
2. List in descending alphabetic order the names of all instructors
3. List in order of the combination of the names and salary of all instructors
4. Find the names of all instructors with salary between \$90,000 and \$100,000
5. Find courses that ran in Fall 2017 or in Spring 2018
6. Find courses that ran in Fall 2017 and in Spring 2018
7. Find courses that ran in Fall 2017 but not in Spring 2018
8. Find courses that ran in Fall 2017 or in Spring 2018, retain all duplications
9. Find all instructors whose salary is null
10. Find all instructors whose salary is not null
11. Null under and, or, with true/false ?

*select null and true
false.
or xx
or xx*

department		instructor		teaches	section	takes	student	course	advisor
<u>dept_name</u>		<u>ID</u>	<u>name</u>	<u>ID</u>	<u>course_id</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	<u>s_id</u>
building		dept_name		course_id	sec_id	course_id	name	dept_name	i_id
budget		salary		semester	year	sec_id	dept_name	credits	
				year		semester	tot_cred		
					building	year			
					room_number	grade			
					time_slot_id				

Basic Query Structure (demo)

- These functions operate on the multiset of values of a column of a relation, and return a value

avg: average value

min: minimum value

max: maximum value

sum: sum of values

count: number of values

select $A_1, A_2, \dots, A_n$  Aggregation function over values over multiple rows

from $r_1, r_2, \dots, r_m$

where P

 every single row

group by columns  New clauses

having condition

 every group

department	instructor	teaches	section	takes	student	course	advisor
<u>dept_name</u> building budget	<u>ID</u> name dept_name salary	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u>	<u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> building room_number time_slot_id	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> grade	<u>ID</u> name dept_name tot_cred	<u>course_id</u> title dept_name credits	<u>s_id</u> <u>i_id</u>

Basic Query Structure (demo)

- These functions operate on the multiset of values of a column of a relation, and return a value

avg: average value

min: minimum value

max: maximum value

sum: sum of values

count: number of values

- Find the highest salary of any instructor. *select max(salary) from ...*
- Find the average salary of instructors in the Computer Science department *select avg(salary) from xx where xx*
- Find the lowest salary of an instructor who have taught a course
- Find the total number of instructors who teach a course in the Spring 2018 semester *select count(distinct ID) from teaches where semester = 'Spring' and year = 2018*
- Find the number of tuples in the *course* relation

department		instructor		teaches	section	takes	student	course	advisor
<u>dept_name</u>		<u>ID</u>	name	<u>ID</u>	<u>course_id</u>	<u>ID</u>	<u>ID</u>	<u>course_id</u>	
building		dept_name		sec_id	semester	course_id	name	title	s_id
		salary		year	building	sec_id	dept_name	dept_name	i_id
					room_number	semester		credits	
					time_slot_id	year	tot_cred		
						grade			

Aggregate – Group By - Having

1. Find the average salary of instructors in each department
select dept_name, avg(salary) from instructor group by dept_name
2. Find the names and average salaries of all departments whose average salary is greater than 42000
having avg(salary) > 42000 from instructor where salary > 7000 group by ...
3. Find the names and average salaries of all departments over instructors whose salary is greater than 7000
4. Find the names and average salaries of all departments whose average salary is greater than 70000
5. Find the average salaries of instructors who have taught a course

*select avg(salary) from instructors where id
in (select id from instructor natural join teaches)*

department		instructor		teaches		section		takes		student		course		advisor	
<u>dept_name</u>		<u>ID</u>	name	<u>ID</u>	<u>course_id</u>	<u>sec_id</u>	<u>semester</u>	<u>ID</u>	<u>course_id</u>	<u>ID</u>	name	<u>course_id</u>	title	<u>s_id</u>	<u>i_id</u>
building		dept_name		<u>sec_id</u>	<u>semester</u>	building	room_number	<u>sec_id</u>	<u>semester</u>	dept_name		dept_name	credits		
budget		salary		<u>year</u>		time_slot_id		<u>year</u>	grade	tot_cred					

Nested Subqueries

- A **subquery** is a **select-from-where** expression that is nested within another query.
- The nesting can be done in the following SQL query

```
select A1, A2, ..., An
from r1, r2, ..., rm
where P
```

" in "

as follows:

- From clause:** r_i can be replaced by any valid subquery
- Where clause:** P can be replaced with an expression of the form:
 $B <\text{operation}> (\text{subquery})$

B is an attribute and $<\text{operation}>$ to be defined later.

- Select clause:**

A_i can be replaced by a subquery that generates a single value.

department	instructor	teaches	section	takes	student	course	advisor
<u>dept_name</u> building budget	<u>ID</u> name dept_name salary	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u>	<u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> building room_number time_slot_id	<u>ID</u> <u>course_id</u> <u>sec_id</u> <u>semester</u> <u>year</u> grade	<u>ID</u> name dept_name tot_cred	<u>course_id</u> title dept_name credits	<u>s_id</u> <u>i_id</u>

```
select * from instructor where name in ('Mort', 'u');
```

- Find names of instructors with salary greater than that of some (at least one) instructor in the Computer Science department.
- ```
select name from instructor where salary > some (select salary
from instructor where dept_name = 'Computer Science')
```

4的some 改成all,  $>$  改  $\geq$

5 的 subqua + where ...

```
count(DISTINCT B.id)
int64
```

|                  |            |                  |                                                                                                                |                  |           |                  |             |
|------------------|------------|------------------|----------------------------------------------------------------------------------------------------------------|------------------|-----------|------------------|-------------|
|                  |            |                  | section                                                                                                        |                  |           |                  |             |
|                  |            | teaches          | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | takes            |           |                  |             |
| department       | instructor | ID               |                                                                                                                | ID               | student   | course           | advisor     |
| <u>dept_name</u> | <u>ID</u>  | <u>course_id</u> |                                                                                                                | <u>course_id</u> | <u>ID</u> | <u>course_id</u> |             |
| building         | name       | <u>sec_id</u>    | year                                                                                                           | <u>sec_id</u>    | name      | title            | <u>s_id</u> |
| budget           | dept_name  | <u>semester</u>  | room_number                                                                                                    | <u>semester</u>  | dept_name | dept_name        | i_id        |
|                  | salary     | <u>year</u>      | time_slot_id                                                                                                   | year             | tot_cred  | credits          |             |
|                  |            |                  |                                                                                                                | grade            |           |                  |             |

# Definition of “some” Clause

- $F <\text{comp}> \text{some } r \Leftrightarrow \exists t \in r \text{ such that } (F <\text{comp}> t)$   
Where  $<\text{comp}>$  can be:  $<, \leq, >, =, \neq$

$(5 < \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline 6 \\ \hline \end{array}) = \text{true}$  (read: 5 < some tuple in the relation)

$(5 < \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{false}$

$(5 = \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{true}$

$(5 \neq \text{some } \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline \end{array}) = \text{true (since } 0 \neq 5)$

$(= \text{some}) \equiv \text{in}$

However,  $(\neq \text{some}) \not\equiv \text{not in}$

# Definition of “all” Clause

- $F \text{ <comp> all } r \Leftrightarrow \forall t \in r (F \text{ <comp> } t)$

$$(5 < \text{all} \begin{array}{|c|} \hline 0 \\ \hline 5 \\ \hline 6 \\ \hline \end{array}) = \text{false}$$

$$(5 < \text{all} \begin{array}{|c|} \hline 6 \\ \hline 10 \\ \hline \end{array}) = \text{true}$$

$$(5 = \text{all} \begin{array}{|c|} \hline 4 \\ \hline 5 \\ \hline \end{array}) = \text{false}$$

$$(5 \neq \text{all} \begin{array}{|c|} \hline 4 \\ \hline 6 \\ \hline \end{array}) = \text{true (since } 5 \neq 4 \text{ and } 5 \neq 6)$$

$(\neq \text{all}) \equiv \text{not in}$

However,  $(= \text{all}) \not\equiv \text{in}$

distinct course\_id.

# Subquery in Where Clause

empty

false

→ text weather empty

not empty

true

- | <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>             |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <i>s_id</i><br><i>i_id</i> |

distinct  
select name from instructor A and instructor B where  
**Query Quest**

A. salary > B. salary and B. dept\_name = 'Comp. Sci.'

```
select B.name from instructor B where exists (select * from instructor A where A.dept_name = 'Comp. Sci.' and B.salary > A.salary);
```

1. Find names of instructors with salary greater than that of some (at least one) instructor in the Computer Science department.

- Use self-join, some, exists, aggregation 集合

2. Find all instructors earning the highest salary (there may be more than one with the same salary). select \* from instructor where salary

- Use self-join, all, exists, aggregation >= all (select salary from

select name from instructor where salary > (select instructor, min(salary) from instructor where dept\_name = 'Comp. Sci.') order by name

| department | instructor                        | teaches                                       | section                                                                            | takes                                                  | student                             | course                                     | advisor      |
|------------|-----------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------|--------------------------------------------|--------------|
| dept_name  | ID<br>name<br>dept_name<br>salary | ID<br>course_id<br>sec_id<br>semester<br>year | course_id<br>sec_id<br>semester<br>year<br>building<br>room_number<br>time_slot_id | ID<br>course_id<br>sec_id<br>semester<br>year<br>grade | ID<br>name<br>dept_name<br>tot_cred | course_id<br>title<br>dept_name<br>credits | s_id<br>i_id |

# Subquery in From and Select Clauses

*select avg\_salary from (select dept\_name, avg(salary) as*

*avg\_salary from instructor  
group by dept\_name)*

*where avg\_sal*

*> 42000*

*(select max(budget from department)*

*where budget = max-budget.value;*

1. Find the average instructors' salaries of those departments where the average salary is greater than \$42,000
2. Find all departments with the maximum budget (with clause)
3. Find all departments where the total salary is greater than the average of the total salary at all departments
4. List all departments along with the number of instructors in each department

```
with totsalarly(dept_name, value) as (select dept_name, sum(salary) from instructor group by dept_name), avgsalary(avgvalue) as (select avg(value) from totsalarly) select * from totsalarly, avgsalary where value > avgsalary.avgvalue;
```

| dept_name  | value    | avgvalue           |
|------------|----------|--------------------|
| Comp. Sci. | 232000.0 | 128285.71428571429 |
| Physics    | 182000.0 | 128285.71428571429 |
| Finance    | 170000.0 | 128285.71428571429 |

section

4.

```
select B.dept_name, (select count(*) from instructor A where A.dept_name = B.dept_name) from department B;
```

| dept_name  | (SELECT count_star() FROM instructor AS A WHERE (A.dept_name = B.dept_name)) |
|------------|------------------------------------------------------------------------------|
| Biology    | 1                                                                            |
| Comp. Sci. | 3                                                                            |
| Elec. Eng. | 1                                                                            |
| Finance    | 2                                                                            |
| History    | 2                                                                            |
| Music      | 1                                                                            |
| Physics    | 2                                                                            |

student

course

ID  
name  
dept\_name  
tot\_cred

course\_id  
title  
dept\_name  
credits

advisor

s\_id  
i\_id

# With Clause

- The **with** clause provides a way of defining a temporary relation whose definition is available only to the query in which the **with** clause occurs.
- Find all departments with the maximum budget

```

with max_budget (value) as
 (select max(budget)
 from department)
select department.name
from department, max_budget
where department.budget = max_budget.value;

```

| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>i_id</u> |

# Query Quest

- Find all departments where the total salary is greater than the average of the total salary at all departments

| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>i_id</u> |

# Modification of the Database

- Deletion of tuples from a given relation.
- Insertion of new tuples into a given relation
- Updating of values in some tuples in a given relation

# Deletion

*tuple*

- Delete all instructors

**delete from** *instructor*

- Delete all instructors from the Finance department

**delete from** *instructor*

**where** *dept\_name* = 'Finance';

- Delete all tuples in the instructor relation for those instructors associated with a department located in the Watson building.

**delete from** *instructor*

**where** *dept name* in (select *dept name*

**from** *department*

**where** *building* = 'Watson');

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                           |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><u><i>i_id</i></u> |

## Deletion (Cont.)

- Delete all instructors whose salary is less than the average salary of instructors

```
delete from instructor
where salary < (select avg (salary)
 from instructor);
```

- Problem: as we delete tuples from *instructor*, the average salary changes
- Solution used in SQL:
  - First, compute **avg** (*salary*) and find all tuples to delete
  - Next, delete all tuples found above (without recomputing **avg** or retesting the tuples)

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                           | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                           |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><i>year</i><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><u><i>i_id</i></u> |

# Insertion

- Add a new tuple to *course*

**insert into** *course*

**values** ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- or equivalently

**insert into** *course* (*course\_id*, *title*, *dept\_name*, *credits*)

**values** ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- Add a new tuple to *student* with *tot\_creds* set to null

**insert into** *student*

**values** ('3003', 'Green', 'Finance', *null*);

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                           |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><u><i>i_id</i></u> |

- Make each student in the Music department who has earned more than 144 credit hours an instructor in the Music department with a salary of \$18,000.

## 批量插入

```
insert into instructor
select ID, name, dept_name, 18000
from student
where dept_name = 'Music' and total_cred > 144;
```

- The **select from where** statement is evaluated fully before any of its results are inserted into the relation.

Otherwise queries like

**insert into table1 select \* from table1**  
would cause problem

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                           |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><u><i>i_id</i></u> |

# Updates

- Give a 5% salary raise to all instructors

```
update instructor
 set salary = salary * 1.05
```

- Give a 5% salary raise to those instructors who earn less than 70000

```
update instructor
 set salary = salary * 1.05
 where salary < 70000;
```

- Give a 5% salary raise to instructors whose salary is less than average

```
update instructor
 set salary = salary * 1.05
 where salary < (select avg (salary)
 from instructor);
```

| department       |           | instructor  |                  | teaches         | section             | takes           | student          | course           | advisor     |
|------------------|-----------|-------------|------------------|-----------------|---------------------|-----------------|------------------|------------------|-------------|
| <u>dept_name</u> | <u>ID</u> | <u>name</u> | <u>dept_name</u> | <u>ID</u>       | <u>course_id</u>    | <u>sec_id</u>   | <u>ID</u>        | <u>course_id</u> |             |
| building         |           |             |                  | <u>sec_id</u>   | <u>semester</u>     | <u>sec_id</u>   | <u>name</u>      | <u>title</u>     | <u>s_id</u> |
|                  |           |             |                  | <u>semester</u> | <u>year</u>         | <u>semester</u> | <u>dept_name</u> | <u>dept_name</u> | <u>i_id</u> |
|                  |           |             |                  | <u>year</u>     | <u>building</u>     | <u>year</u>     | <u>tot_cred</u>  | <u>credits</u>   |             |
|                  |           |             |                  |                 | <u>room_number</u>  | <u>grade</u>    |                  |                  |             |
|                  |           |             |                  |                 | <u>time_slot_id</u> |                 |                  |                  |             |

# Updates (Cont.)

- Increase salaries of instructors whose salary is over \$100,000 by 3%, and all others by a 5%

- Write two **update** statements:

```
update instructor
 set salary = salary * 1.03
 where salary > 100000;
update instructor
 set salary = salary * 1.05
 where salary <= 100000;
```

order! 不能反。  
否则有些老师的涨2次

- The order is important
- Can be done better using the **case** statement (next slide)

| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>i_id</u> |

# Case Statement for Conditional Updates

- Same query as before but with case statement

**update** *instructor*

**set** *salary* = **case**

**when** *salary* <= 100000 **then** *salary* \* 1.05

**else** *salary* \* 1.03

**end**

| <i>department</i>       | <i>instructor</i> | <i>teaches</i>          | <i>section</i>          | <i>takes</i>            | <i>student</i>   | <i>course</i>           | <i>advisor</i>     |
|-------------------------|-------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|--------------------|
| <u><i>dept_name</i></u> | <u><i>ID</i></u>  | <u><i>ID</i></u>        | <u><i>course_id</i></u> | <u><i>ID</i></u>        | <u><i>ID</i></u> | <u><i>course_id</i></u> | <u><i>s_id</i></u> |
| <i>building</i>         | <i>name</i>       | <u><i>course_id</i></u> | <u><i>sec_id</i></u>    | <u><i>course_id</i></u> | <i>name</i>      | <i>title</i>            | <u><i>i_id</i></u> |
| <i>budget</i>           | <i>dept_name</i>  | <u><i>sec_id</i></u>    | <u><i>semester</i></u>  | <u><i>sec_id</i></u>    | <i>dept_name</i> | <i>dept_name</i>        |                    |
|                         | <i>salary</i>     | <u><i>semester</i></u>  | <i>building</i>         | <u><i>semester</i></u>  | <i>tot_cred</i>  | <i>credits</i>          |                    |
|                         |                   | <u><i>year</i></u>      | <i>room_number</i>      | <u><i>year</i></u>      |                  |                         |                    |
|                         |                   |                         | <i>time_slot_id</i>     | <i>grade</i>            |                  |                         |                    |

value.

# Updates with Scalar Subqueries

- Recompute and update tot\_creds value for all students

**update** *student* *S*

**set** *tot\_cred* = (**select** **sum**(*credits*)

**from** *takes*, *course*

**where** *takes.course\_id* = *course.course\_id* **and**

*S.ID* = *takes.ID* **and**

*takes.grade* <> 'F' **and**

*takes.grade* **is not null**);

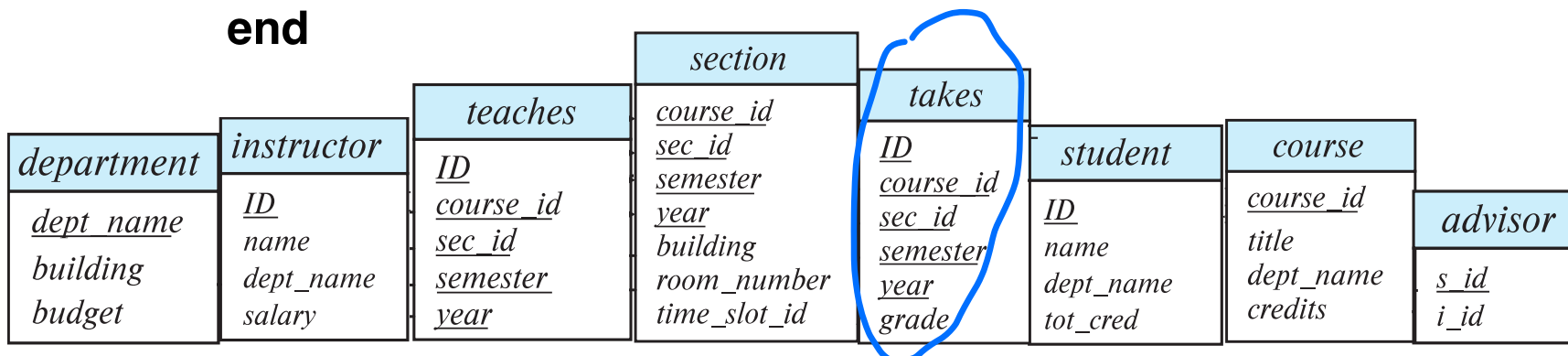
- Sets *tot\_creds* to null for students who have not taken any course
- Instead of **sum**(*credits*), use:

**case**

**when** **sum**(*credits*) **is not null** **then** **sum**(*credits*)

**else** 0

**end**



FIN

Any questions?

# Join types

- Three forms of outer join:
  - Natural left outer join *course* ⋈ *prereq*

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> |
|------------------|--------------|------------------|----------------|
| BIO-301          | Genetics     | Biology          | 4              |
| CS-190           | Game Design  | Comp. Sci.       | 4              |
| CS-315           | Robotics     | Comp. Sci.       | 3              |

| <i>course_id</i> | <i>prereq_id</i> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| CS-190           | CS-101           |
| CS-347           | CS-101           |

| <i>department</i> |                  | <i>teaches</i>  |                  | <i>section</i>      | <i>takes</i>     | <i>student</i>   | <i>course</i>    | <i>advisor</i> |
|-------------------|------------------|-----------------|------------------|---------------------|------------------|------------------|------------------|----------------|
| <i>dept_name</i>  | <i>ID</i>        | <i>ID</i>       | <i>course_id</i> | <i>sec_id</i>       | <i>ID</i>        | <i>ID</i>        | <i>course_id</i> |                |
| <i>building</i>   | <i>name</i>      | <i>sec_id</i>   | <i>sec_id</i>    | <i>semester</i>     | <i>course_id</i> | <i>name</i>      | <i>title</i>     | <i>s_id</i>    |
| <i>budget</i>     | <i>dept_name</i> | <i>semester</i> | <i>semester</i>  | <i>building</i>     | <i>sec_id</i>    | <i>dept_name</i> | <i>dept_name</i> | <i>i_id</i>    |
|                   | <i>salary</i>    | <i>year</i>     | <i>year</i>      | <i>room_number</i>  | <i>semester</i>  | <i>tot_cred</i>  | <i>credits</i>   |                |
|                   |                  |                 |                  | <i>time_slot_id</i> | <i>year</i>      |                  |                  |                |
|                   |                  |                 |                  |                     | <i>grade</i>     |                  |                  |                |

# Join types

- Three forms of outer join:
  - Natural left outer join  $course \bowtie_{\text{prereq}}$

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> |
|------------------|--------------|------------------|----------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           |
| CS-315           | Robotics     | Comp. Sci.       | 3              | <i>null</i>      |

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> |
|------------------|--------------|------------------|----------------|
| BIO-301          | Genetics     | Biology          | 4              |
| CS-190           | Game Design  | Comp. Sci.       | 4              |
| CS-315           | Robotics     | Comp. Sci.       | 3              |

| <i>course_id</i> | <i>prereq_id</i> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| CS-190           | CS-101           |
| CS-347           | CS-101           |

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                           | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>             |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><i>year</i><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <i>s_id</i><br><i>i_id</i> |

# Join types

## Three forms of outer join:

- Natural left outer join *course* ⋈ *prereq*
- Natural right outer join *course* ⋈ *prereq*
- Natural full outer join *course* ⋈ *prereq*

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> |
|------------------|--------------|------------------|----------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           |
| CS-315           | Robotics     | Comp. Sci.       | 3              | <i>null</i>      |

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> |
|------------------|--------------|------------------|----------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           |
| CS-347           | <i>null</i>  | <i>null</i>      | <i>null</i>    | CS-101           |

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> |
|------------------|--------------|------------------|----------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           |
| CS-315           | Robotics     | Comp. Sci.       | 3              | <i>null</i>      |
| CS-347           | <i>null</i>  | <i>null</i>      | <i>null</i>    | CS-101           |

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> |
|------------------|--------------|------------------|----------------|
| BIO-301          | Genetics     | Biology          | 4              |
| CS-190           | Game Design  | Comp. Sci.       | 4              |
| CS-315           | Robotics     | Comp. Sci.       | 3              |

| <i>course_id</i> | <i>prereq_id</i> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| CS-190           | CS-101           |
| CS-347           | CS-101           |

| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>             |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <i>s_id</i><br><i>i_id</i> |

# Joined Relations – Examples

- course natural right outer join prereq

| course_id | title       | dept_name  | credits | prereq_id |
|-----------|-------------|------------|---------|-----------|
| BIO-301   | Genetics    | Biology    | 4       | BIO-101   |
| CS-190    | Game Design | Comp. Sci. | 4       | CS-101    |
| CS-347    | null        | null       | null    | CS-101    |

- course full outer join prereq using (course\_id)

| course_id | title       | dept_name  | credits | prereq_id |
|-----------|-------------|------------|---------|-----------|
| BIO-301   | Genetics    | Biology    | 4       | BIO-101   |
| CS-190    | Game Design | Comp. Sci. | 4       | CS-101    |
| CS-315    | Robotics    | Comp. Sci. | 3       | null      |
| CS-347    | null        | null       | null    | CS-101    |

| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor             |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|---------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br>i_id |

for nature join, keep the same value

## Joined Relations – Examples

- course **inner join** prereq on  
course.course\_id = prereq.course\_id

remain all

| course_id | title       | dept_name  | credits | prereq_id | course_id |
|-----------|-------------|------------|---------|-----------|-----------|
| BIO-301   | Genetics    | Biology    | 4       | BIO-101   | BIO-301   |
| CS-190    | Game Design | Comp. Sci. | 4       | CS-101    | CS-190    |

- What is the difference between the above, and a natural join?
- course **left outer join** prereq on  
course.course\_id = prereq.course\_id

| course_id | title       | dept_name  | credits | prereq_id | course_id |
|-----------|-------------|------------|---------|-----------|-----------|
| BIO-301   | Genetics    | Biology    | 4       | BIO-101   | BIO-301   |
| CS-190    | Game Design | Comp. Sci. | 4       | CS-101    | CS-190    |
| CS-315    | Robotics    | Comp. Sci. | 3       | null      | null      |

|                                        |                                          | section                                                                          |                                                                                                                |                                                                                           |                                            |                                                   |                                       |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|---------------------------------------|
| department                             | instructor                               | teaches                                                                          |                                                                                                                | takes                                                                                     | student                                    | course                                            |                                       |
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | advisor<br><u>s_id</u><br><u>i_id</u> |

# View

Virtual

- A **view** provides a mechanism to hide certain data from the view of certain users.

- Consider a person who needs to know an instructors name and department, but not the salary. This person should see a relation described, in SQL, by

```
select ID, name, dept_name
from instructor
```

- create view v as** < query expression >

- A view of instructors without their salary
- Find all instructors in the Biology department
- Create a view of department salary totals

view 可嵌套  
不真实存在  
每次执行, 更新从表

| department       |           | teaches   |                  | section          | takes     |                  | student   | course           | advisor     |             |
|------------------|-----------|-----------|------------------|------------------|-----------|------------------|-----------|------------------|-------------|-------------|
| <u>dept_name</u> | <u>ID</u> | <u>ID</u> | <u>course_id</u> | <u>course_id</u> | <u>ID</u> | <u>course_id</u> | <u>ID</u> | <u>course_id</u> | <u>s_id</u> | <u>i_id</u> |
| building         | name      | sec_id    | sec_id           | semester         | sec_id    | sec_id           | name      | title            |             |             |
|                  | dept_name | semester  | semester         | year             | semester  | year             | dept_name | dept_name        |             |             |
| budget           | salary    | year      | year             | building         | year      | grade            | tot_cred  | credits          |             |             |
|                  |           |           |                  | room_number      |           |                  |           |                  |             |             |
|                  |           |           |                  | time_slot_id     |           |                  |           |                  |             |             |

## Define Views using other Views

- create view **physics\_fall\_2017** as  
 select course.course\_id, sec\_id, building, room\_number  
 from course, section  
 where course.course\_id = section.course\_id  
 and course.dept\_name = 'Physics'  
 and section.semester = 'Fall'  
 and section.year = '2017';
- create view **physics\_fall\_2017\_watson** as  
 select course\_id, room\_number  
 from **physics\_fall\_2017**  
 where building = 'Watson';

| department       |  | instructor |  | teaches   | section          | takes     | student   | course           | advisor |
|------------------|--|------------|--|-----------|------------------|-----------|-----------|------------------|---------|
| <u>dept_name</u> |  | <u>ID</u>  |  | <u>ID</u> | <u>course_id</u> | <u>ID</u> | <u>ID</u> | <u>course_id</u> |         |
| building         |  | name       |  | course_id | sec_id           | course_id | name      | title            | s_id    |
|                  |  | dept_name  |  | sec_id    | semester         | sec_id    | dept_name | dept_name        | i_id    |
|                  |  | salary     |  | semester  | year             | semester  | tot_cred  | credits          |         |
|                  |  |            |  | year      | building         | year      |           |                  |         |
|                  |  |            |  |           | room_number      | grade     |           |                  |         |
|                  |  |            |  |           | time_slot_id     |           |           |                  |         |

# Join

| Join types              |
|-------------------------|
| <b>inner join</b>       |
| <b>left outer join</b>  |
| <b>right outer join</b> |
| <b>full outer join</b>  |

| Join conditions                         |
|-----------------------------------------|
| <b>natural</b>                          |
| <b>on</b> < predicate >                 |
| <b>using</b> ( $A_1, A_2, \dots, A_n$ ) |

| <u>course_id</u> | <u>title</u> | <u>dept_name</u> | <u>credits</u> |
|------------------|--------------|------------------|----------------|
| BIO-301          | Genetics     | Biology          | 4              |
| CS-190           | Game Design  | Comp. Sci.       | 4              |
| <b>CS-315</b>    | Robotics     | Comp. Sci.       | 3              |

| <u>course_id</u> | <u>prereq_id</u> |
|------------------|------------------|
| BIO-301          | BIO-101          |
| CS-190           | CS-101           |
| <b>CS-347</b>    | CS-101           |

| department       |                 | instructor       |               | teaches   |                  | section       |                 | takes       |                 | student            |                     | course    |                  | advisor       |                 |             |              |           |                  |               |                 |             |             |                  |                 |                  |              |                  |                |             |             |
|------------------|-----------------|------------------|---------------|-----------|------------------|---------------|-----------------|-------------|-----------------|--------------------|---------------------|-----------|------------------|---------------|-----------------|-------------|--------------|-----------|------------------|---------------|-----------------|-------------|-------------|------------------|-----------------|------------------|--------------|------------------|----------------|-------------|-------------|
| <u>dept_name</u> | <u>building</u> | <u>dept_name</u> | <u>budget</u> | <u>ID</u> | <u>course_id</u> | <u>sec_id</u> | <u>semester</u> | <u>year</u> | <u>building</u> | <u>room_number</u> | <u>time_slot_id</u> | <u>ID</u> | <u>course_id</u> | <u>sec_id</u> | <u>semester</u> | <u>year</u> | <u>grade</u> | <u>ID</u> | <u>course_id</u> | <u>sec_id</u> | <u>semester</u> | <u>year</u> | <u>name</u> | <u>dept_name</u> | <u>tot_cred</u> | <u>course_id</u> | <u>title</u> | <u>dept_name</u> | <u>credits</u> | <u>s_id</u> | <u>i_id</u> |

# Joined Relations – Examples

- *course* **inner join** *prereq* **on** *course.course\_id = prereq.course\_id*

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> | <i>course_id</i> |
|------------------|--------------|------------------|----------------|------------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          | BIO-301          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           | CS-190           |

- What is the difference between the above, and a natural join?
- *course* **left outer join** *prereq* **on** *course.course\_id = prereq.course\_id*

| <i>course_id</i> | <i>title</i> | <i>dept_name</i> | <i>credits</i> | <i>prereq_id</i> | <i>course_id</i> |
|------------------|--------------|------------------|----------------|------------------|------------------|
| BIO-301          | Genetics     | Biology          | 4              | BIO-101          | BIO-301          |
| CS-190           | Game Design  | Comp. Sci.       | 4              | CS-101           | CS-190           |
| CS-315           | Robotics     | Comp. Sci.       | 3              | <i>null</i>      | <i>null</i>      |

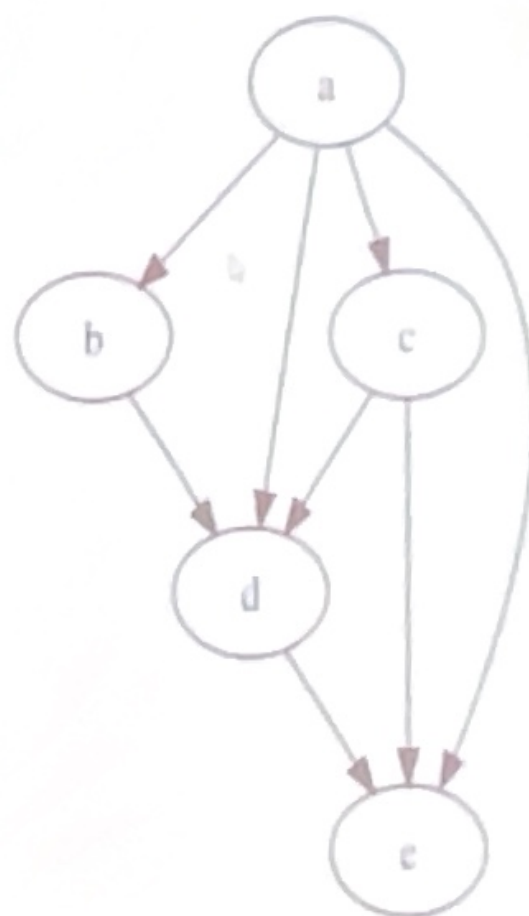
| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                                  | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                    |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><i>i_id</i> |

# Query Quest

1. Display a list of all instructors, showing their ID, name, and the number of sections that they have taught. Make sure to show the number of sections as 0 for instructors who have not taught any section. Your query should use an outer join, and should not use scalar subqueries.
2. Outer join expressions can be computed in SQL without using the SQL outer join operation. To illustrate this fact, show how to rewrite each of the following SQL queries without using the outer join expression.
  1. **select \* from student natural left outer join takes**
  2. (hard) **select \* from student natural full outer join takes**
    1. Select id, name, count(coalesce(course\_id, sec\_id, semester, year)) from instructor natural left outer join teaches group by id;

| department       |  | instructor |      | teaches   |                  | section       |                 | takes     |                  | student   |      | course           |       | advisor     |             |
|------------------|--|------------|------|-----------|------------------|---------------|-----------------|-----------|------------------|-----------|------|------------------|-------|-------------|-------------|
| <u>dept_name</u> |  | <u>ID</u>  | name | <u>ID</u> | <u>course_id</u> | <u>sec_id</u> | <u>semester</u> | <u>ID</u> | <u>course_id</u> | <u>ID</u> | name | <u>course_id</u> | title | <u>s_id</u> | <u>i_id</u> |
| building         |  | dept_name  |      | sec_id    |                  | year          |                 | sec_id    |                  | dept_name |      | dept_name        |       |             |             |
| budget           |  | salary     |      | semester  |                  | building      |                 | semester  |                  | tot_cred  |      | credits          |       |             |             |
|                  |  |            |      | year      |                  | room_number   |                 | year      |                  |           |      |                  |       |             |             |
|                  |  |            |      |           |                  | time_slot_id  |                 | grade     |                  |           |      |                  |       |             |             |

# Dependency Graph of Views should be a DAG

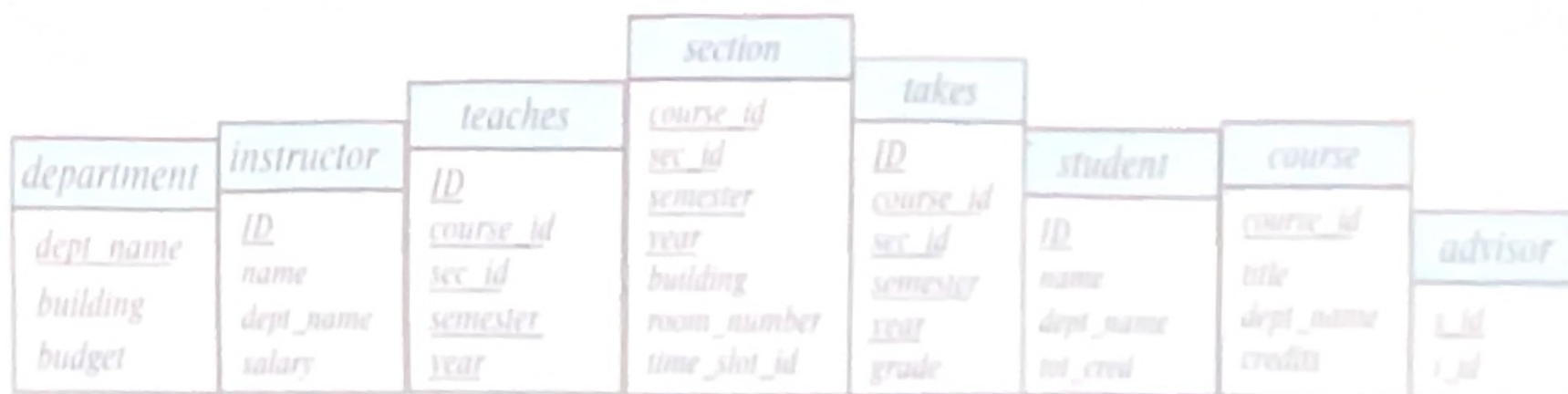


| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>t_id</u> |

# Integrity Constraints

完整性约束

- A checking account must have a balance greater than \$10,000.00
- A salary of a bank employee must be at least \$4.00 an hour
- A customer must have a (non-null) phone number



# Constraints on a Single Relation

- not null
  - name* varchar(20) not null
  - budget* numeric(12,2) not null
- primary key
- unique (  $A_1, A_2, \dots, A_m$  )
  - Candidate keys are permitted to be null (in contrast to primary keys).
- check (P), where P is a predicate (read DDL.sql)

**create table** department

(dept\_name          varchar(20), building                  varchar(15),  
 budget              numeric(12,2) **check** (budget > 0),  
**primary key** (dept\_name));

| department       |  | instructor |  | teaches   |  | section          |  | takes     |  | student   |  | course           |  | advisor     |  |
|------------------|--|------------|--|-----------|--|------------------|--|-----------|--|-----------|--|------------------|--|-------------|--|
| <u>dept_name</u> |  | <u>ID</u>  |  | <u>ID</u> |  | <u>course_id</u> |  | <u>ID</u> |  | <u>ID</u> |  | <u>course_id</u> |  | <u>i_id</u> |  |
| building         |  | name       |  | course_id |  | sec_id           |  | course_id |  | name      |  | title            |  | i_id        |  |
| budget           |  | dept_name  |  | sec_id    |  | year             |  | sec_id    |  | dept_name |  | dept_name        |  | i_id        |  |
|                  |  | salary     |  | semester  |  | building         |  | semester  |  | tot_cred  |  | credit           |  |             |  |
|                  |  |            |  | year      |  | room_number      |  | year      |  |           |  |                  |  |             |  |
|                  |  |            |  |           |  | time_slot_id     |  | grade     |  |           |  |                  |  |             |  |

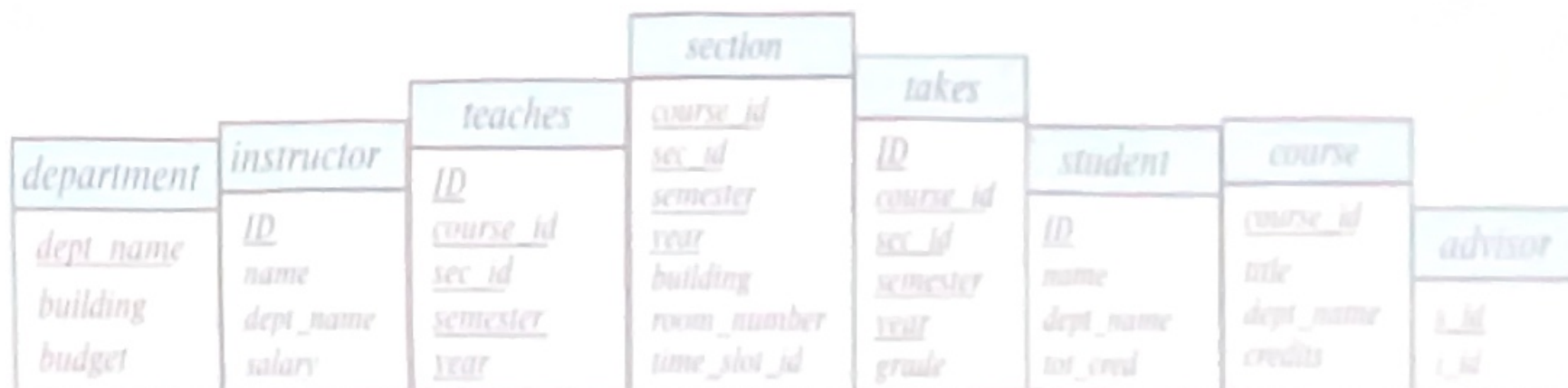
# Integrity Constraint Violation

- create table *person* (  
     *ID* char(10),  
     name char(40),  
     mother char(10),  
     father char(10),  
     primary key *ID*,  
     foreign key *father* references *person*,  
     foreign key *mother* references *person*)
- How to insert a tuple without causing constraint violation?

| department                             | instructor                               | teaches                                                                          | section                                                                                                        | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>i_id</u> |

# Built-in Data Types in SQL

- **date**: Example: **date** '2005-7-27'
- **time**: Example: **time** '09:00:30'      **time** '09:00:30.75'
- **Timestamp** Example: **timestamp** '2005-7-27 09:00:30.75'
- **interval**: Subtracting a date/time/timestamp value from another gives an interval value
- Large objects (photos, videos, CAD files, etc.) are stored as a *large object*:
  - **blob**: binary large object
  - **clob/text**: character large object
  - When a query returns a large object, a pointer is returned rather than the large object itself.



# Index Creation

- We create an index with the **create index** command  
**create index <name> on <relation-name> (attribute);**

|                  |            |                  |                  |                  |           |                  |             |
|------------------|------------|------------------|------------------|------------------|-----------|------------------|-------------|
|                  |            |                  | section          |                  |           |                  |             |
| department       | instructor | teaches          |                  | takes            | student   | course           | advisor     |
| <u>dept_name</u> | <u>ID</u>  | <u>ID</u>        | <u>course_id</u> | <u>ID</u>        | <u>ID</u> | <u>course_id</u> |             |
| building         | name       | <u>course_id</u> | <u>sec_id</u>    | <u>course_id</u> | name      | title            | <u>s_id</u> |
|                  | dept_name  | <u>sec_id</u>    | <u>semester</u>  | <u>sec_id</u>    | dept_name | dept_name        | <u>i_id</u> |
| budget           | salary     | <u>semester</u>  | <u>year</u>      | <u>semester</u>  | tot_cred  | credits          |             |
|                  |            | <u>year</u>      | building         | <u>year</u>      |           |                  |             |
|                  |            |                  | room_number      | grade            |           |                  |             |
|                  |            |                  | time_slot_id     |                  |           |                  |             |

# Index Creation

- We create an index with the **create index** command  

```
create index <name> on <relation-name> (attribute);
```
- **create table** *student* (*ID* varchar (5), *name* varchar (20) not null, *dept\_name* varchar (20), *tot\_cred* numeric (3,0) default 0, primary key (*ID*))
- **create index** *studentID\_index* on *student*(*ID*)
- The query:  

```
select *
from student
where ID = '12345'
```

## Demo: EXPLAIN QUERY PLAN

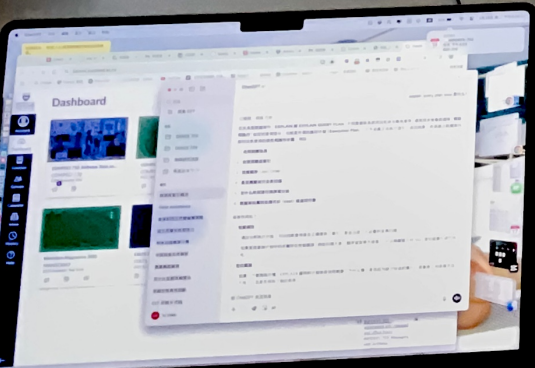
| <i>department</i>                                           | <i>instructor</i>                                                    | <i>teaches</i>                                                                                                      | <i>section</i>                                                                                                                                           | <i>takes</i>                                                                                                                        | <i>student</i>                                                         | <i>course</i>                                                                 | <i>advisor</i>                           |
|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|
| <u><i>dept_name</i></u><br><i>building</i><br><i>budget</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>salary</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u> | <u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><i>year</i><br><i>building</i><br><i>room_number</i><br><i>time_slot_id</i> | <u><i>ID</i></u><br><u><i>course_id</i></u><br><u><i>sec_id</i></u><br><u><i>semester</i></u><br><u><i>year</i></u><br><i>grade</i> | <u><i>ID</i></u><br><i>name</i><br><i>dept_name</i><br><i>tot_cred</i> | <u><i>course_id</i></u><br><i>title</i><br><i>dept_name</i><br><i>credits</i> | <u><i>s_id</i></u><br><u><i>i_id</i></u> |

```

sqlite> select * from student;
+-----+-----+-----+-----+
| ID | name | dept_name | tot_cred |
+-----+-----+-----+-----+
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier		56
45678	Levy		46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow		0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120
+-----+-----+-----+-----+

sqlite> explain query plan select * from student where id = '54321';
QUERY PLAN
|--SEARCH student USING INDEX sqlite_autoindex_student_1 (ID=?)
sqlite>

```



Inc

- V
- c
- d
- p
- c
- T

Demo

depar  
degr  
build  
hang

# Transactions

事項、仕事

- The transaction must end with one of the following statements:
  - **Commit work.**
  - **Rollback work.**
- Atomic transaction
- Isolation from concurrent transactions
- ACID
  - Consistency and Durability.
- Demo: transaction

| department                             | instructor                               | teaches                                                                          | section                                                                                                 | takes                                                                                     | student                                    | course                                            | advisor                    |
|----------------------------------------|------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|----------------------------|
| <u>dept_name</u><br>building<br>budget | <u>ID</u><br>name<br>dept_name<br>salary | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u> | <u>course_id</u><br><u>sec_id</u><br><u>semester</u><br>year<br>building<br>room_number<br>time_slot_id | <u>ID</u><br><u>course_id</u><br><u>sec_id</u><br><u>semester</u><br><u>year</u><br>grade | <u>ID</u><br>name<br>dept_name<br>tot_cred | <u>course_id</u><br>title<br>dept_name<br>credits | <u>s_id</u><br><u>i_id</u> |